



UGANDA NATIONAL EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

P720/1

TECHNICAL DRAWING

(Geometrical Drawing)

**SCORING GUIDE
FOR THE SAMPLE PAPER**

ITEM 1:

To attain the highest level of achievement, the candidate is required to:

- Correctly laying out the drawing sheet with all required details, including a frame/border line and a title block with clearly printed information.
- Follow the correct procedures to analyse, interpret, and prepare the required information and working drawings using appropriate symbols and conventions.
- Appropriate use of line types, scales, dimensions, and annotations on the drawings presented.
- Produce all required working drawings that can be used in solving a problem that is in the scenario (Fit for purpose)

Possible Responses

1. Correctly Drawing frame and title block printed with:
 - Candidate's identification (name, personal and random no.)
 - Title of drawing
 - Scale
 - Date
 - Client
2. Producing required scaled drawings using symbols and conventions, choosing and using the appropriate scale and using suitable line types, annotations and dimensions. The drawings include
 - Front elevations
 - Plan
 - development following
 - ✓ steps followed include;
 - Drawing axes
 - Drawing octagon
 - Drawing oblique cone circle in the plan
 - Drawing base pipe
 - Octagon in front view
 - Dividing the circle
 - Projecting to elevation
 - Determining the cone apex
 - Projecting cone points to the apex
 - Completing the elevation
 - Projecting from elevation to complete plan
 - Curves to complete plan
 - Plotting curve

- Triangulation lines on elevation
- Triangulation on plan
- True length of the curve
- True length of transition piece
- True length of oblique cone
- Plotting trap
- Joining points for the trap
- Plotting pipe
- joining points for pipe

3. Product that is fit for purpose should have:

- Accurate dimensions and measurements – all sizes are correct and clearly shown.
- Appropriate scale – drawings are drawn to a stated and suitable scale.
- Correct use of symbols and line types – standard conventions are properly applied.
- Clear annotations and dimensions – information is well labelled and easy to understand.
- Neat and well-organised presentation – includes proper layout, border lines, and a complete title block

Item 2

To attain the highest level of achievement, the candidate is required to:

- Correctly laying out the drawing sheet with all required details, including a frame/border line and a title block with clearly printed information.
- Follow the correct procedures to analyse, interpret, and prepare the required information and working drawings using appropriate symbols and conventions.
- Appropriate use of line types, scales, dimensions, and annotations on the drawings presented.
- Produce all required working drawings that can be used in solving a problem that is in the scenario (Fit for purpose)

Possible Responses

1. Correctly Drawing frame and title block printed with:

- Candidate's identification (name, personal and random no.)
- Title of drawing
- Scale used
- Date
- Client

2. Producing required scaled drawings using symbols and conventions, choosing and using the appropriate scale and using suitable line types, annotations and dimensions. Steps/procedures include

- Line presentation of data
- Space diagram (bow's notation)
- Projection lines
- Load line
- Joining points to load line
- Bending moments
- Completing polar diagram
- Determining reactions
- Shear force diagram

3. Product that is fit for purpose should have:

- Accurate dimensions and measurements – all sizes are correct and clearly shown.
- Appropriate scale – drawings are drawn to a stated and suitable scale.
- Correct use of symbols and line types – standard conventions are properly applied.
- Clear annotations and dimensions – information is well labelled and easy to understand.
- Neat and well-organised presentation – includes proper layout, border lines, and a complete title block